Yashas K

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Objective

2nd-year engineering student with proven Rust systems engineering experience. Seeking a Software Engineer internship at Disney Streaming Zurich to apply succinct data structure and low-level memory-layout work to high-throughput content indexing and video streaming infrastructure. Independent research record includes a published stress-tested binary retrieval graph with measured 94% storage reduction.

Core Skills

Languages	Rust (1.99, release builds, LTO), Python (3.12), C, shell scripting
Systems	Low-level memory layout, bit-packed data structures, CPU cache-aware design, succinct tries (LOUDS), LZ77-style compression
Benchmarking	Real measurement methodology (no extrapolation), `std::time::Instant`, `fs::metadata`, gzip/xz/bzip2 comparison
Tools	cargo, rustup, git, GitHub Actions, LaTeX (Tectonic), matplotlib
Research	ORCID-registered, IEEE conference paper format, reproducibility scripts, honest limitations documentation

Featured Project: AXIOM — Binary Substring Graph for Zero-Error Retrieval

github.com/Yashas-K-Gangatkar/axiom-3d · Rust 1.99 · 3 binaries, 8-page IEEE paper, 5 charts, MIT-style license · **October 2026**

Designed architectural primitives applicable to high-throughput content indexing and low-latency prefix search systems.

- **Built a three-layer binary retrieval graph in Rust**: V1 prefix trie, V2 LZ77 macro graph with byte-level hash chains, V3.1 LOUDS succinct trie with rank/select navigation. Each layer stress-tested on real Project Gutenberg text at 1, 10, and 100 MB.
- **Achieved 94% storage reduction** by replacing pointer-based trie nodes (26 bytes/node) with LOUDS bit-packed encoding (4.5 bits/node). On 10 MB of text: V1 = 2.05 GB, V3.1 = 121.5 MB.
- **Measured 0% hallucination rate** across all corpus sizes and all three layers, verified against 5 adversarial fake queries per size. The exact-match property is structural—no probability in the retrieval path—and directly applicable as a verification gate behind FAISS-class ANN indexes in RAG pipelines.
- **Benchmarked against standard compressors** (gzip -9, xz -9, bzip2 -9) on identical corpora. Honestly documented that V2 is 38× worse than `xz` on compression ratio due to missing entropy coding, and that substring search is $O(N{\cdot}L)$ rather than $O(L)$ without a suffix-tree upgrade.
- **Shipped research-grade tooling**: reproducibility scripts, GitHub Actions CI, JSON output of every measured number, IEEE conference LaTeX paper with 14 references. Every claim in the paper is traceable to a measurement in the repository.

Honest limitations documented in the paper (recruiters value engineers who know where their code breaks):

- V1 bloats 212× on disk (solved by V3.1 LOUDS, 94% reduction)
- Substring search is $O(N{\cdot}L)$, not $O(L)$—suffix-tree upgrade is future work
- V2 has no entropy coding layer (38× gap to `xz`)
- Single-threaded build (no Rayon parallelism yet)

Education

Bachelor of Engineering, Computer Science 2024 – 2028 (expected)
Engineering College, Bangalore, India 2nd year (current)

- Independent research focus: low-level data structures, succinct encoding, retrieval systems
- Built AXIOM outside coursework as a self-directed engineering study
- ORCID registered: 0009-0005-8928-5341

Other Work

- **YK Compressor** (2024–2025): earlier Rust prototype of spatial pattern matching for genomic data. Public on GitHub. Superseded by AXIOM; the lessons learned informed AXIOM's honest measurement methodology.
- **Open-source contribution**: All AXIOM code is open-source under PolyForm Noncommercial 1.0.0. Reproducibility is a first-class concern: every number in the paper traces to a JSON file in the repo.

Research Output

- **AXIOM: A Binary Substring Graph for Zero-Error Retrieval** (October 2026). IEEE conference format preprint, 8 pages, 14 references. PDF on GitHub. Target venues: IEEE Big Data 2027, ACM CIKM 2027, NeurIPS workshops on efficient ML.
- **ETH Student Summer Research Fellowship** — applying November 2026 for summer 2027 placement. Will propose work on suffix-tree upgrade for AXIOM to achieve $O(L)$ substring search.

Why Disney Streaming Zurich

Disney Streaming's work on video encoding, recommendation infrastructure, and high-throughput content pipelines aligns directly with the systems-engineering skills I developed building AXIOM:

- LOUDS succinct tries (94% storage reduction in AXIOM V3.1) apply directly to video frame indexing at scale, where pointer-based indexes become memory-bound.
- LZ77 macro chains (AXIOM V2) are the same family of techniques used in modern video codecs for inter-frame deduplication.
- The honest trade-offs I documented in AXIOM—storage vs latency, build time vs query time, exact vs approximate retrieval—are the same trade-offs a streaming infrastructure team navigates daily.

I am applying for a Software Engineer internship (3–6 months) or a full-time SWE 1 role. I am specifically interested in the encoding pipeline, content indexing, and recommendation infrastructure teams.

Availability

- **Internship**: Available May–August 2027 (3 months) or any 3–6 month window in 2027
- **Full-time SWE 1**: Available from July 2028 (post-graduation)
- **Work permit**: Will require Swiss work permit sponsorship; open to Talent Visa (`Nr. 1`) pathway
- **Relocation**: Willing to relocate to Zurich; German language learning in progress

References available on request. All claims in this resume are verifiable in the AXIOM GitHub repository. github.com/Yashas-K-Gangatkar/axiom-3d